



Cancers of the eye

Abhilasha Maheshwari¹ · Paul T. Finger¹

© Springer Science+Business Media, LLC, part of Springer Nature 2018

Abstract

Ocular cancers are unique among the diseases of the eye, threatening both vision and life. In most cases, the diagnosis can be made utilizing a careful clinical history and specialized ocular examination. Eye cancer diagnosis relies heavily on imaging techniques such as high-frequency ultrasound, fluorescein angiography, anterior and posterior segment optical coherence tomography, computed tomography (CT), and magnetic resonance imaging (MRI). Once the diagnosis is established, treatment decisions depend on the tumor's location, size, local extension, patterns of growth, and secondary complications. Treatment options include observation, local resection, chemotherapy (topical, intravenous, intra-arterial, or intravitreal), and radiation (ophthalmic plaque or external beam). Enucleation or exenteration is only employed if these eye- and vision-sparing treatments are not possible. The core of this comprehensive review is a consecutive series of the most common ocular tumor of each structure of the eye, anterior to posterior, including basal cell carcinoma of the eyelid, squamous conjunctival neoplasia, choroidal melanoma, retinoblastoma, ocular adnexal lymphoma, and metastatic orbital tumors.

Keywords Basal cell carcinoma · Ocular surface squamous neoplasia · Choroidal melanoma · Retinoblastoma · Ocular adnexal lymphoma · Metastasis

1 Introduction

Malignant ophthalmic tumors are rare and diverse, so their diagnosis can be quite complex. They are unique among the diseases of the eye and threaten both vision and life. Compared to most ophthalmic diseases, management of patients with a malignant ophthalmic tumor presents additional challenges. Therapy is directed to save life, vision, and the eye (in that order). Already requiring special expertise and equipment, ophthalmic oncology is rapidly advancing due to accelerating progress in tumor biology, pharmacology, and instrumentation. The care of most eye cancer patients often requires a multidisciplinary team, comprising ocular oncologists, general oncologists, radiotherapists, pathologists, psychologists, and other specialists.

This comprehensive review is intended as an overview of ocular oncology with reference to the most common types of malignant ocular tumor of each structure of the eye.

✉ Paul T. Finger
pfinger@eyecancer.com

Abhilasha Maheshwari
dimension.abhi@gmail.com

¹ The New York Eye Cancer Center, Suite 5B, 115 East 61st Street, New York, NY 10065, USA

2 Eyelid

Basal cell carcinoma (BCC) is the most common malignant eyelid cancer (90%) [1]. It occurs in sixth to eighth decade of life, more commonly in men [1]. Risk factors for periocular BCC include ultraviolet (UV) radiation, local and systemic immune dysfunction, and prior exposure to ionizing radiation [1]. It is postulated to arise from the basal cells of the epidermis and most commonly arises from the lower eyelid, medial to the inner canthus and upper eyelid (in that order) [1].

BCC is a slowly growing, firm, indurated, nodular lesion which can be either pigmented or non-pigmented. The nodular form commonly has pearly margins and a central crater. The epithelial surface of the tumor is usually smooth (multinodular) with fine, telangiectatic, visible tumor vessels. On clinical examination, the notable features are patchy crusts, papules, loss of eyelashes, and malpositioned lid margins. Less common BCC presentations include and may be overlapping: nodular, nodulo-ulcerative, pigmented, cystic, and infiltrating (morpheaform or sclerotic). Palpation of preauricular, submandibular, and cervical lymph nodes for regional spread is mandatory.

BCC is locally invasive and can invade the orbit and sinuses. Imaging is, hence, necessary to rule out orbital extension. T1-weighted contrast-enhanced fat-suppressed MRI

scans are the modality of choice for demonstrating a soft tissue mass or infiltration. Computed tomographic (CT) bone window and ultrasound imaging may also be used. The American Joint Committee on Cancer Staging (AJCC) eighth edition classification for eyelid carcinomas is given in Table 1 [2].

The differential diagnosis for BCC includes sebaceous gland carcinoma, squamous cell carcinoma, keratoacanthoma, trichoepithelioma, desmoplastic trichoepithelioma, and metastatic carcinoma.

Diagnostic histopathological workup includes exfoliative cytology, incisional, or excisional biopsies [3]. Histopathologically, BCC is characterized by a proliferation of cells with oval nuclei and scanty cytoplasm that form infiltrative nests. At the periphery of these nests, cells are arranged radially, and this pattern is called “palisading.” Necrosis is a common finding in BCC.

The main treatment modality for BCC is complete surgical excision of the lesion with microscopically clear margins (Mohs micrographic surgery or traditional surgical excision), followed by eyelid reconstruction [1, 4]. Very small and superficial tumors (< 2 mm) can be treated by vaporization with carbon dioxide laser or cryotherapy [5–7]. Both these techniques have more risk of scarring. Small tumors (3–4 mm) can be removed by pentagonal wedge excision and direct lid reconstruction. Medium-sized (7–8 mm) lesions must be reconstructed with local flaps (e.g., Tenzel’s, Mustarde’s, or glabellar). Large tumor resection (> 10 mm) typically requires reconstruction with Hughes or Cutler-Beard techniques [8–10]. Radiotherapy can be used as an adjuvant to exenteration in cases with orbital invasion and/or BCC unresponsive to alternative therapies [11].

Table 1 AJCC eighth edition classification for eyelid carcinomas [2]

	TNM category	Description
Primary tumor (T)	TX	Primary tumor cannot be assessed
	T0	No evidence of primary tumor
	Tis	Carcinoma <i>in situ</i>
	T1	Tumor ≤ 10 mm in greatest dimension
	T1a	Tumor does not invade tarsal plate or eyelid margin
	T1b	Tumor invades the tarsal plate or eyelid margin
	T1c	Tumor involves full thickness of the eyelid
	T2	Tumor > 10 or ≤ 20 mm in greatest dimension
	T2a	Tumor does not invade tarsal plate or eyelid margin
	T2b	Tumor invades the tarsal plate or eyelid margin
	T2c	Tumor involves full thickness of the eyelid
	T3	Tumor > 20 or ≤ 30 mm in greatest dimension
	T3a	Tumor does not invade tarsal plate or eyelid margin
	T3b	Tumor invades the tarsal plate or eyelid margin
	T3c	Tumor involves full thickness of the eyelid
	T4	Any eyelid tumor that invades adjacent ocular, orbital, or facial structures
	T4a	Tumor invades ocular or intraorbital structures
	T4b	Tumor invades the bony walls if the orbit or extends to the paranasal sinuses or invades the lacrimal sac/nasolacrimal duct or brain
Regional lymph nodes (N)	NX	Regional lymph nodes cannot be assessed
	N0	No evidence of lymph node involvement
	N1	Metastasis in single ipsilateral regional lymph node, ≤ 3 cm in greatest dimension
	N1a	Metastasis in single ipsilateral lymph node based on clinical evaluation or imaging findings
	N1b	Metastasis in single ipsilateral lymph node based on lymph node biopsy
	N2	Metastasis in single ipsilateral lymph node, > 3 cm in greatest dimension, or in bilateral or contralateral lymph nodes
	N2a	Metastasis documented based on clinical evaluation or imaging findings
	N2b	Metastasis documented based on lymph node biopsy
Distant metastasis (M)	M0	No distant metastasis
	M1	Distant metastasis

A new class of immune response modifier (topical imiquimod cream) showed the potential to provide topical treatment of BCC for which excision would cause significant aesthetic and functional compromise [12]. According to deMacedo et al., daily application, 5 days per week for 8–16 weeks, can be applied until the lesion becomes undetectable [13]. In this study, the 3-year histological clearance rate was 81.8% for lesions > 10 mm and 100% for lesions < 10 mm. Adverse effects of topical treatment included conjunctivitis, keratitis, foreign body sensation, lacrimation, decreased vision, and ectropion [13]. Vismodegib is a hedgehog pathway inhibitor that was approved in 2012 for treatment of locally advanced or metastatic BCC [14]. It produced significant clinical responses and can avoid exenteration in select patients, while causing manageable drug-related adverse effects [14, 15].

In treatment of BCC, poor prognostic factors are younger age (< 35 years), large tumor size (> 10 mm), location near the canthi, morpheaform or infiltrative growth pattern, deep invasion, and recurrent tumor [1, 11]. Long-term clinical follow-up is important because two-thirds of recurrences appear within 3 years of treatment and 18% between 5 and 10 years [1]. Recurrent tumors are more aggressive and difficult to manage [1]. BCC rarely metastasizes (0.01%). However, when they do, the frequent sites of metastasis are lymph nodes, lungs, bone, and liver. Risk factors for metastasis are multiple recurrences, aggressive histopathological appearance, and orbital/sinus invasion [16].

3 Conjunctiva

Ocular surface squamous “malignant” neoplasia (OSSN) is the spectrum of conjunctival lesions that includes dysplasia, carcinoma *in situ*, and squamous cell carcinoma (SCC) [17]. The incidence of OSSN was 0.3 per million in the USA [18]. A study from Brisbane, Australia, found the incidence to be 1.9 per million in Australia [19]. It is more common in Caucasian men and those with immune deficiency [17]. For example, reports suggest that those infected with HIV are at a relatively high risk for OSSN and their tumors are recalcitrant to treatment [2]. The other important factors are exposure to ultraviolet radiation and human papilloma virus (HPV). These factors can alter the cells at the limbus, resulting in the formation of OSSN [17].

The most common symptoms are foreign body sensation, ocular redness, and irritation. Vision is not affected until the center of the cornea is involved. The interpalpebral fissure at the limbus is the most common primary location. The lesion is slightly elevated, relatively well demarcated, pearly to reddish white with the presence of feeder vessels, intrinsic “corkscrew” vascularity, and episcleral fixation. The clinical types of OSSN are gelatinous, papilliform, and leukoplakic. Tumors

can present with each type or a combination thereof. In addition, OSSN can either be primarily nodular or diffuse. Palpation for regional lymph node extension should be done in every patient.

The differential diagnosis of OSSN includes pinguecula, pterygium, and squamous papilloma [20]. For histopathological diagnosis, exfoliative and impression cytology can be done [21]. The specificity of these techniques is low as tumor edges and deep margins are not determined [21]. Recent publications have described the use of high-frequency ultrasound in estimating the depth of invasion or involvement of the angle [22]. The 8th edition AJCC classification for conjunctival carcinoma is shown in Table 2 [23].

Surgical excision is the traditional method of treatment for most small tumors [20]. Its success depends on the involvement of the peripheral and deep margins. To avoid recurrence, we recommend excising the entire visible tumor and an additional 2 mm safety margin. Cryotherapy should be employed to sterilize the deep scleral or corneal margins and an additional 2–3 mm of treated conjunctival margins (beyond the lines of resection) by double freeze-thaw technique [24]. “Finger-tip” applicators have been developed to produce homogeneous freezing over large surface areas, allowing for more uniform treatment of tissue [24]. OSSN patients initially treated at The New York Eye Cancer Center were found to have recurrence rates as low as 4% [25]. Small conjunctival defects (< 3 mm) can be covered by direct suturing of the conjunctiva whereas large defects may need to be covered with free or pedicle-based conjunctival auto-grafts or donor amniotic membrane.

Larger (> 3 mm), multifocal, or aggressive tumors typically require combinations of resection, adjuvant cryotherapy, and topical chemotherapy. After the use of high-frequency ultrasonographic or radiographic imaging to rule out intraocular invasion, primary therapy involves topical chemotherapy. In order of preference at The New York Eye Cancer Center, we use interferon alpha 2B, 5-fluorouracil, or mitomycin. Thus, we start with the medication least likely to induce side effects and sequentially progress to more aggressive therapies [26]. Surgery (excision with cryotherapy) has become the second line of treatment for recalcitrant cases. Methods of topical chemotherapy vary widely [25–30].

Interferon alpha 2B has been used, either *via* intrascleral injection or *via* drops, with complete clinical resolution of the tumors [27]. Recently, subconjunctival ranibizumab and external beam radiation therapy using photons, electrons, or protons have been proposed as palliative treatments for non-responsive carcinomas of the eyelids and conjunctiva [28, 29]. As a last resort, orbital exenteration with adjuvant external beam radiation therapy may be required [30].

Intraocular invasion has been reported to occur in up to 5% of cases [31]. Patients at risk are typically older, those with SCC located near the corneoscleral limbus, the

Table 2 AJCC eighth edition classification for conjunctival carcinomas [23]

	TNM category	Description
Primary tumor (T)	TX	Primary tumor cannot be assessed
	T0	No evidence of primary tumor
	Tis	Carcinoma <i>in situ</i>
	T1	Tumor ≤ 5 mm in greatest dimension invades through the conjunctival basement membrane without invasion of adjacent structures
	T2	Tumor > 5 mm in greatest dimension invades through the conjunctival basement membrane without invasion of adjacent structures
	T3	Tumor invades adjacent structures (excluding the orbit)
	T4	Tumor invades the orbit with or without further extension
	T4a	Tumor invades orbital soft tissue without bone invasion
	T4b	Tumor invades bone
	T4c	Tumor invades adjacent paranasal sinus
	T4d	Tumor invades brain
Regional lymph nodes (N)	NX	Regional lymph nodes cannot be assessed
	N0	No regional lymph node metastasis
	N1	Regional lymph node metastasis
Distant metastasis (M)	M0	No distant metastasis
	M1	Distant metastasis

immunodeficient, and those with one or more recurrences after surgical excision [31]. Metastasis of conjunctival SCC is rare in developed countries, but more likely in AJCC T3 and T4-sized tumors [32, 33]. The main cause of metastasis is delay in diagnosis and treatment. Sites of metastasis include preauricular, submandibular, and cervical lymph nodes, parotid gland, lungs, and bones.

4 Retina

Retinoblastoma (RB) represents 3% of all childhood cancers and is the most common intraocular malignancy of childhood [34]. The reported worldwide incidence is 1 in 15,000 to 23,000 live births [35]. The tumor occurs bilaterally in 30–40% of cases. Of the newly diagnosed cases of retinoblastoma, only 6% are familial whereas 94% are sporadic [34]. The average age at diagnosis is 12 months (bilateral) and 24 months (unilateral) [34]. There is no racial or gender predilection.

The most common clinical presentation is leukocoria (50%) followed by strabismus (20%) [36]. Other presentations include decreased vision, painful red eye, hyphema, or proptosis [36]. At presentation, a thorough clinical history with family history should be obtained.

Every patient should undergo ultrasound imaging to look for size of the tumor, status of vitreous, and highly reflective intratumoral calcification which is diagnostic for retinoblastoma. Examination under general anesthesia involves slit lamp examination, intraocular pressure measurement, and dilated

fundoscopy. Fundus photography should be done at every visit for comparison. Radiographic imaging (MRI or CT) is performed to evaluate optic nerve and extraocular extension (particularly in cases with atypical manifestations like proptosis, anterior chamber involvement, and neovascular glaucoma) [37]. On clinical examination, four types of growth patterns of retinoblastoma can be noted: exophytic, endophytic, mixed, and diffuse infiltrating. Patients should be classified to determine the type of treatment required. The older international classification and the new 8th edition AJCC TNMH Classification of retinoblastoma are shown in Tables 3 and 4 respectively [38, 39].

The primary goals of management of retinoblastoma are to save life, preserve vision and the eye. Following a detailed evaluation of the child with retinoblastoma, the family should be counselled regarding the diagnosis, its implications on the life, eye and vision, the treatment plan, costs, and the need for a long-term follow-up.

History and clinical examination can typically be used to rule out the differential diagnoses: Coats' disease, persistent fetal vasculature, retinopathy of prematurity, congenital cataract, coloboma, astrocytic hamartoma, endogenous endophthalmitis, toxocariasis, familial exudative vitreoretinopathy, and juvenile xanthogranuloma [40].

Retinoblastoma biopsy is usually contraindicated due to risk of extraocular tumor extension in an open eye [41]. The various modalities of treatment are intravenous chemotherapy, transpupillary thermotherapy, cryotherapy, plaque brachytherapy, external beam radiotherapy, intra-arterial chemotherapy, intravitreal chemotherapy, and enucleation [38, 42–44].

Table 3 The international classification of retinoblastoma [38]

Group	Philadelphia version	Los Angeles version
A	RB < 3 mm	RB < 3 mm, at least 3 mm from the foveola, and 1.5 mm from optic nerve. No seeding
B	RB > 3 mm or macular location or juxtapapillary location (< 1.5 mm to disc) or SRF present	Eyes with no vitreous or subretinal seeding and retinal tumors of any size or location not included in group A. Small cuff of subretinal fluid $\leq$ 5 mm from tumor margin
C	RB with SRS < 3 mm from RB or VS $\leq$ 3 mm from RB	Eyes with focal vitreous or subretinal seeding and discrete tumor of any size or location. Seeding must be local, fine, and limited to be theoretically treatable with a radioactive plaque. Up to 1 quadrant subretinal fluid may be present.
D	RB with SRS > 3 mm from RB or VS > 3 mm from RB	Eyes with diffuse vitreous or subretinal seeding and/or massive, non-discrete endophytic, or exophytic disease. Seeding more extensive than group C. Retinal detachment > 1 quadrant
E	RB with size > 50% of globe or neovascular glaucoma or opaque media or invasion of optic nerve, choroid, sclera, orbit, anterior chamber	Massive RB with anatomic or functional destruction of the eye with one or more of the following: Neovascular glaucoma Massive intraocular hemorrhage Aseptic orbital cellulitis Tumor anterior to anterior vitreous face Tumor touching lens Diffuse infiltrating tumor Phthisis or pre-phthisis

RB retinoblastoma, SRF subretinal fluid, SRS subretinal seeds, VS vitreous seeds

Intravenous chemotherapy is used as a primary treatment, for presumed microscopic metastasis in high-risk RB patients with massive choroidal invasion, extrascleral, or intraneural tumor extension [38]. It is also used for treatment of metastatic retinoblastoma. Chemotherapeutic agents include vincristine, carboplatin, and etoposide [38]. It is important to note that systemic chemotherapy is toxic to bone marrow and may cause infertility and secondary non-ocular malignancies [38].

A new treatment “intra-arterial chemotherapy” requires an interventional radiology specialist. Chemotherapy is injected directly into the ophthalmic artery with a goal to minimize the side effects of systemic chemotherapy. Chemotherapeutic agents used are typically melphalan, carboplatin, and topotecan [38]. Average of three sessions per eye at 4-week intervals is given. Vitreous and anterior chamber tumor seeding and hemorrhage are predictive for treatment failure. Residual RB can be treated with laser, cryotherapy, or plaque brachytherapy. Risks include thromboemboli in the optic nerve, disappearance of high-risk histopathologic features (down-staging), and untreated occult metastases.

A newer treatment, “intravitreal chemotherapy,” is yet to be defined. However, it appears to be important as second-line therapy for recurrent vitreous RB seeds [38]. Munier et al. studied 23 patients with heavily treated retinoblastoma with recurrent vitreous seeds, treated with 20–30 mg melphalan on a weekly basis and noted globe retention rate of 87% [45].

Transpupillary thermotherapy with infrared 810 nm or argon 532 nm is used as combination therapy for small ($\leq$ 2.5–3.0 mm) or multifocal tumors [46]. Argon laser is more focal

and associated with less laser scar enlargement over time. Cryotherapy was an important modality, but currently is rarely used in favor of other focal therapies for posterior tumors. However, cryotherapy remains important for treatment of small (2 mm in thickness and 4 mm in basal diameter) equatorial and peripheral tumors (less amenable to laser) [34].

Retinoblastoma is radiation sensitive. Primary brachytherapy is used for smaller, anterior, solitary tumors [47]. It is used for salvage in cases of residual or recurrent tumors. Isotopes used are ^{125}I , ^{103}Pd , and ^{106}Ru . Radioactive plaque is sewn onto the eye (sclera) over the tumor and a dose of 40–45 Gray (Gy) is given over 2 to 5 days [42].

External beam radiation is rarely used due to local side effects and risk of secondary cancers [48]. In addition to permanent dry eye, retinopathy, and disturbed growth of orbital bones (and teeth), it can cause secondary malignancies (e.g., sarcomas) [48]. It can be used for ocular salvage in cases of chemotherapy failure; tumor recurrence in one-eyed patient; treatment of orbital tumor invasion (after enucleation) and for intracranial RB [43].

Enucleation is the only method that completely removes the primary tumor [44]. It is used for very large tumors (involving > 75% of vitreous volume); those with extra scleral extension or intraneural tumor invasion; and massive choroidal, vitreous, subretinal, or anterior chamber tumor seeding [49]. It is also used as the method of choice if the tumor relapses after eye-sparing treatment [49]. Enucleation provides histopathologic features. Of those, poor histopathologic prognostic factors are massive “> 3-mm” uveal invasion,

Table 4 AJCC eighth edition classification for retinoblastoma [39]

	TNM category	Description
Primary tumor (T)	TX	Unknown evidence of intraocular tumor
	T0	No evidence of intraocular tumor
	Tis	Carcinoma <i>in situ</i>
	T1	Intraretinal tumor(s) with subretinal fluid ≤ 5 mm from the base of any tumor
	T1a	Tumor < 3 mm and further than 1.5 mm from disc and fovea
	T1b	Tumor > 3 mm and closer than 1.5 mm from disc and fovea
	T2	Intraocular tumor(s) with retinal detachment, vitreous seeding, or subretinal seeding
	T2a	Subretinal fluid > 5 mm from the base of any tumor
	T2b	Vitreous seeding and/or subretinal seeding
	T3	Advanced intraocular tumor(s)
	T3a	Phthisis or pre-phthisis bulbi
	T3b	Tumor invasion of choroid, pars plana, ciliary body, lens, zonules, iris, or anterior chamber
	T3c	Raised intraocular pressure with neovascularization and/or buphthalmos
	T3d	Hyphema and/or massive vitreous hemorrhage
	T3e	Aseptic orbital cellulitis
	T4	Extraocular tumor(s) involving orbit, including optic nerve
	T4a	Radiologic evidence of retrobulbar optic nerve involvement or thickening of optic nerve or involvement of orbital tissues
	T4b	Extraocular tumor clinically evident with proptosis and/or orbital mass
Regional lymph nodes (N)	NX	Regional lymph nodes cannot be assessed
	N0	No regional lymph node involvement
	N1	Evidence of preauricular, submandibular, and cervical lymph node involvement
Distant metastasis (M)	M0	No sign or symptom of intracranial or distant metastasis
	M1	Distant metastasis without microscopic confirmation
	M1a	Tumor(s) involving any distant site (e.g., bone marrow, liver) on clinical or radiologic tests
	M1b	Tumor involving the CNS or radiologic imaging (not including trilateral retinoblastoma)
Heritable trait	HX	Unknown or insufficient evidence of a constitutional RB1 gene mutation
	H0	Normal RB1 alleles in blood tested with demonstrated high-sensitivity assays
	H1	Bilateral retinoblastoma, retinoblastoma with an intracranial primitive neuroectodermal tumor (i.e., trilateral retinoblastoma), patient with family history of retinoblastoma, or molecular definition of a constitutional RB1 gene mutation

extra scleral tumor extension, optic nerve invasion, and anterior chamber extension [50].

Post treatment monitoring for local and systemic recurrence includes close monitoring by the eye cancer specialist and pediatric oncologist, physical examination, and blood tests (especially for those treated with chemotherapy or radiation therapy). Ophthalmic examinations are necessary to rule out the development of subsequent retinoblastoma in the same or other eye, chorioretinal toxicity after chemotherapy, radiation retinopathy, and optic neuropathy [45, 46]. Those with only family history of retinoblastoma (but without the disease)

need to be screened at least until the age of 28 months. Bilaterality, delays in diagnosis, germline mutation, RB1 allele, and immunocompromised patient are important predictors of metastatic retinoblastoma [51].

In developed countries, few patients with retinoblastoma die from their disease [52]. Most will retain vision in one or both eyes and lead fairly normal lives. Unfortunately, more than 70% of children in less developed countries do not survive retinoblastoma [52]. Therefore, the most important advances in survival will be gained by providing care to these children.

5 Choroid

Choroidal melanomas are the most common primary intraocular malignancy in adults, accounting for 5% of all melanomas [53]. More common in Caucasians, the Collaborative Ocular Melanoma Study (COMS) found the mean age at diagnosis to be 60 years [54, 55]. The incidence of ocular melanomas for whites in the USA [Surveillance and Epidemiology and End Result (SEER) program] was 0.69 per 100,000 person-years for males and 0.59 for females [56, 57]. Though an association between UV light exposure and uveal melanoma is controversial, there exists literature suggesting that those with occupational exposure (e.g., welders, dentists) and the population beneath the Australian ozone hole are at greater risk [53, 58–60].

Choroidal melanomas are located posterior to the ciliary body. Often asymptomatic, in developed nations, they are most commonly found during ophthalmic examination. When symptomatic, they can induce early changes in vision and/or symptoms of “flashing lights” due to tumor-induced exudative retinal detachment or metamorphopsia due to subfoveal tumor. In an effort to help ophthalmic healthcare professionals diagnose choroidal melanoma, Dr. Finger created the mnemonic “MOST” [61]. It is based on the most common characteristics of choroidal melanoma.

As a choroidal Melanoma invades and destroys the overlying retinal pigment epithelium, accumulations of Orange pigment (lipofuscin and melanolipofuscin) can be visualized on the surface of the melanoma. Meanwhile, incontinent tumor blood vessels leak serum beneath the retina, exudative Subretinal fluid. Lastly, choroidal melanomas are typically ≥ 2 mm Thick. Though not all choroidal melanomas exhibit all three findings, **MOST** is diagnostic when all three are present [61].

At The New York Eye Cancer Center, office-based imaging techniques typically include fundus photography (for serial monitoring of the margins and surface of tumor), fluorescein angiography (FA) (to evaluate tumor-related vasculopathy, cystoid macular edema, and exudative subretinal fluid), fundus auto fluorescence (FAF) imaging (to detect orange pigment lipofuscin), optical coherence tomography (OCT) (to detect subretinal fluid, drusenoid retinal pigment epithelial detachments), and ultrasound imaging (to assess tumor size, internal reflectivity, and extrascleral extension). Few cases may require magnetic resonance imaging [MRI] or computed tomography [CT] in the event of suspected extrascleral extension [62]. The additional imaging is necessary in cases with slow diminution of vision, proptosis, optic nerve congestion, choroidal elevation, and rapid progression of lesion [63]. Positron emission tomography/computed tomography (PET-CT) is typically performed to stage the melanoma (rule out metastasis) [64].

Biopsy is controversial and rarely required for diagnosis because discohesive tumor cells are liberated within the eye and are known to exit through the sclerostomy into the orbit

[65]. Despite this, no statistically significant medical evidence exists that those cells metastasize [65]. Accepted biopsy risks include vitreous hemorrhage, retinal tear, retinal detachment, endophthalmitis, cataract, and epiretinal membrane formation [65]. Indications for uveal tumor biopsy include atypical tumors, metastatic tumors with no known primary, patient request, and for genetic/molecular analysis [65].

According to the 8th edition of the American Joint Committee on Cancer-International Union Against Cancer classification system, uveal melanoma size can be classified as T1, T2, T3, and T4 [66]. This helps in both staging and for prediction of metastatic risk (Fig. 1).

In 1985, COMS established iodine-125 (^{125}I) as the most common and widely used radionuclide for treatment of choroidal melanoma [67]. Various other radioactive materials have been used for plaque brachytherapy like ^{125}I , ^{60}Co , ^{103}Pd , and ^{106}Ru [68–70]. Low-energy plaques (^{125}I or ^{103}Pd) are bowl-shaped devices constructed to house rice-sized radioactive sources (or seeds) [71]. Ruthenium106 (^{106}Ru) plaques are solid disc-shaped beta-sources where the radionuclide is covered by a 0.1-mm-thick layer of silver at its episcleral aperture [72]. The plaque is temporarily sutured to the sclera overlying the tumor, left in place for 5 to 7 days, and then surgically removed [71]. The first choice for treatment of choroidal melanoma at The New York Eye Cancer center is ^{103}Pd . Many studies have now shown that for almost all melanomas, ^{103}Pd episcleral plaque brachytherapy photons are more rapidly absorbed in tissue than those derived from ^{125}I [73].

Proton beam radiotherapy and stereotactic radiotherapy enable a high dose of radiation to be aimed at the uveal melanoma irrespective of the tumor’s size, shape, or location [74, 75]. As compared to external beam radiation (EBRT) and proton beam therapy, plaque brachytherapy has been associated with fewer anterior segment complications (e.g., cataract, neovascular glaucoma, keratitis, iridocyclitis, and dry eye) [74, 75]. Other methods like transpupillary thermotherapy, photodynamic therapy, and cryotherapy have been tried but suffer from poor efficacy (local control) [76–78]. The use of ^{103}Pd plaques typically results in less irradiation of critical normal ocular structures (e.g. fovea, optic disc, lens) and greater eye and vision preservation.

At the New York Eye Cancer Center, we currently perform primary enucleation in $< 6\%$ of all patients presenting with uveal melanoma. Primary enucleation for uveal melanoma is performed only when other methods are considered unlikely to conserve the eye and useful vision. In cases with tumors filling more than two-thirds of vitreous cavity, extraocular extension or optic nerve extension primary enucleation is the treatment of choice.

5.1 Post-plaque surveillance

At The New York Eye Cancer Center, patients are followed up with periodic ophthalmic examinations including visual

Fig. 1 AJCC eighth edition classification for ciliary body and choroidal melanoma based on thickness and diameter [66]

Melanoma thickness (mm)	AJCC T-Category						
> 15	4	4	4	4	4	4	4
12.1 - 15.0	3	3	3	3	3	4	4
9.1 - 12.0	3	3	3	3	3	3	4
6.1 - 9.0	2	2	2	2	3	3	4
3.1 - 6.0	1	1	1	2	2	3	4
< or = 3.0	1	1	1	1	2	2	4
	< or = 3.0	3.1 - 6.0	6.1 - 9.0	9.1 - 12.0	12.1 - 15.0	15.1 - 18.0	> 18.0
	Melanoma basal diameter (mm)						

acuity, fundus photography, ultrasound imaging, and OCT every 3 to 4 months after brachytherapy. Fluorescein angiography is performed every 6 months. Tumor height (by ultrasoundography) was compared at each visit. Post-treatment surveillance for metastasis involves abdominal radiographic imaging every 6 months for the first 5 years and then every year thereafter.

Bad prognostic factors for uveal melanoma are large tumor diameter (> 16 mm), anterior tumor location with ciliary body involvement, extrascleral extension, diffuse melanoma subtype, older age, male gender, and faster-growing tumor [79]. Metastasis is typically first discovered in the liver, being involved in 95% of cases followed by lung 24%, bone 16%, and skin 11% [80]. The literature also contains reports of hepatic artery chemoembolization, hepatic perfusion, or radiofrequency ablation for treatment of metastatic melanoma [56, 73, 74].

6 Ocular adnexal lymphoma

The ocular adnexa include the eyelids, conjunctiva, lacrimal apparatus, and orbital soft tissue. One percent of all lymphomas and approximately 8% of all extranodal lymphomas arise in the ocular adnexa and their incidence is increasing [81]. Ocular adnexal lymphoma (OAL) represents the malignant end of the spectrum of ocular adnexal lymphoproliferative disease and more commonly affects those in 6th to 7th decade [82, 83]. It affects both genders, with a slight female predilection [82, 83]. Bilaterality is present in 10–20% of cases [82, 83]. The frequency of involvement in decreasing order is eyelid (67%), orbit (35%), and conjunctiva (20%) [82, 83]. Postulated etiologies are clonal proliferation of B or T lymphocytes or because of antigenic inflammation or autoimmune disorders or associated with microbial pathogens (*Helicobacter pylori*, *Chlamydia pneumonia*, *Chlamydia psittaci*) [61].

Majority of OAL are of non-Hodgkin's B cell type. OAL belongs to one of the five subtypes: most common is extra nodal marginal zone (EMZL or MALT lymphoma), follicular lymphoma (FL), diffuse large B cell lymphoma (DLBCL), mantle cell lymphoma (MCL), and lymphoplasmacytic lymphoma (LPL) [84]. The Ann Arbor system is widely used for clinical staging of nodal lymphomas but is not ideally suited to

extra nodal diseases such as OAL. The AJCC eighth edition classification for ocular adnexal lymphomas offers site-specific staging and is given in Table 5 [85].

Patients present with a visible mass lesion, exophthalmos, pain, or diplopia, but may also be asymptomatic [83]. Signs may be site-specific and help in diagnosis. Conjunctival lesions typically have a salmon or flesh-pink color. In the orbit, lacrimal gland, and eyelid, the lymphoma may present as an unseen mass, firm on palpation with presence of exophthalmos, and decreased retropulsion of the globe with or without presence of ptosis [83]. If the lacrimal glands are involved, dry eye is present. Secondary ptosis may occur.

Ophthalmic examination at our center includes slit lamp biomicroscopy of all conjunctival surfaces with eversion of the upper and lower eyelids, exophthalmometry, visual field testing, color vision testing, ocular motility evaluation, inspection, and palpation of eyelids and orbit. Intraocular pressure measurements and findings on dilated ophthalmoscopy may indicate compressive ocular disease [86].

Contrast-enhanced CT or MRI (T1-weighted images dark tumor compared to orbital fat) of the orbits show enhancing lesions which can be discrete or diffuse [87]. Lymphoid lesions typically mold around or against adjacent structures such as the globe or bony orbit [87].

Differential diagnosis includes benign lymphoproliferative lesions, epithelial tumors, melanocytic tumors, and inflammatory/infectious lesions and any mass including orbital metastases [85].

At our center, tumors are biopsied for diagnosis by pathology. Tumor cell growth fraction is assessed by counting the number of tumor cells with clear nuclear positivity for Ki-67 per 5 × 100 tumor cells using the × 40 objective [61, 88]. Pathologic analysis can identify lymphomas but cannot reliably differentiate its types. However, immunohistochemistry can differentiate between various types of OAL [61, 88]. Laboratory evaluation includes a full blood count, hepatic enzymes, serum lactate dehydrogenase (LDH), and bone marrow aspiration and biopsy [61]. A whole-body PET/CT scan should also be obtained to stage the patient and thus rule out other sites of involvement [89, 90].

Low dose EBRT is the gold standard for local disease because OALs are typically radiation sensitive [91]. EBRT is

Table 5 AJCC eighth edition classification for ocular adnexal lymphomas [85]

	TNM category	Description
Primary tumor (T)	TX	Lymphoma extent not specified
	T0	No evidence of lymphoma
	T1	Lymphoma involving the conjunctiva alone without eyelid or orbital involvement
	T2	Lymphoma with orbital involvement with or without conjunctival involvement
	T3	Lymphoma with pre-septal eyelid involvement with or without orbital involvement and with or without conjunctival involvement
	T4	Orbital adnexal lymphoma and extra orbital lymphoma extending beyond the orbit to adjacent structures, such as bone, maxillofacial sinuses, and brain
Regional lymph nodes (N)	NX	Involvement of lymph nodes not assessed
	N0	No evidence of lymph node involvement
	N1	Involvement of lymph node region or regions draining the ocular adnexal structures and superior to the mediastinum (preauricular, parotid, submandibular, and cervical nodes)
	N1a	Involvement of single lymph node region superior to mediastinum
	N1b	Involvement of two or more lymph node regions, superior to mediastinum
	N2	Involvement of lymph node regions of the mediastinum
	N3	Diffuse or disseminated involvement of peripheral and central lymph node regions
Distant metastasis (M)	M0	No evidence of involvement of other extra nodal sites
	M1a	Noncontiguous involvement of tissues or organs external to the ocular adnexa (e.g., parotid gland, submandibular glands, lung, liver, kidney, breast)
	M1b	Lymphomatous involvement of bone marrow
	M1c	Both M1a and M1b involvement

administered as 30 Gy (range 20–40) in 180–200 cGy daily fractions. Analysis of the radiation dose-response relationship of EMZL revealed that 5-year local tumor control rates were 81% with doses below 30 Gy but 100% with doses higher than that [91]. Patient must be monitored for radiation side effects (dry eye, cataract, radiation retinopathy, optic neuropathy) post-treatment [61].

ENMZL (or MALT) conjunctival tumors are less aggressive and associated with low morbidity. Once a pathologic diagnosis is established, most can be observed prior to systemic workup [61]. Cryotherapy has been thought to debulk but not eliminate tumor [84]. It may have an application in patients with conjunctival OAL unable to receive other treatment modalities [84]. Low-dose EBRT (20–26 Gy) in 180–200 cGy fractions is typically curative and rarely causes retinopathy [91].

Combined chemotherapy regimens [cyclophosphamide, doxorubicin, vincristine, and prednisone (CHOP)] are used for more aggressive subtypes such as diffuse large B cell or mantle cell lymphoma [61]. It is less commonly used due to systemic side effects [61].

Immunotherapy with rituximab has been recently approved for relapsed or refractory low-grade B cell NHL and follicular

lymphoma [92–94]. It is given in cycles of four weekly infusions. The local control and systemic efficacy are unknown [92–94].

Antibiotic therapy with doxycycline 100 mg BID PO for 3 weeks can result in a complete or partial response in patients with chronic infection with *C. psittaci* and *H. pylori* [95, 96].

Post treatment assessment for metastasis includes systemic examination with a hematologist-oncologist and periodic monitoring with physical examination, CT abdomen, chest and pelvis or whole-body PET/CT, and blood investigations (complete blood count, LDH) [97–99].

Among OAL, EMZL has a quantitatively better prognosis than other tumor types regarding spread and lymphoma-related death, though the risk ratio is similar among the milder forms EMZL, LPCL, and FCL. The mortality ranges were EMZL 0–20%, DLBCL 25–75%, FL 20–37%, MCL 38–100%, and LPL 14–100% [82, 100, 101].

7 Metastasis

The ophthalmologists diagnosing and treating orbital metastases play a vital role in the diagnosis and management of such patients. This is because up to 35% of patients with orbital

metastasis have no known history of primary tumor at the time of ophthalmologic examination [102, 103]. The frequency of metastatic tumors among orbital mass lesions ranges from 2.5 to 10% [104, 105]. The average age depends on the type of primary tumor: 59 years for breast, 58 years for lung, 70 years for prostate, and 45 years for malignant melanoma [106]. The most frequent primary sites are breast, lung, prostate, and malignant melanoma of the skin [107–109]. In males, the most frequent primary site is lung, and in females, it is breast [106, 109]. Other uncommon primary sites are gastrointestinal tract, salivary glands, kidney, thyroid, urinary bladder, genital organs, and soft tissue tumors [110–117].

History taking is the most important part in the medical evaluation of patients with presumed metastatic disease. This is because primary tumors in other body sites treated several years prior can lead to orbital manifestations. A pathognomonic feature of orbital metastasis is its progressive course associated with motor and sensory deficits. Individual symptoms show specific tissue involvement within the orbit. Tumors infiltrating fat and extraocular muscles produce proptosis [104, 105, 107, 109]. Muscle involvement also produces diplopia. Pain is associated with periosteal and bone involvement. Pulsations are due to bone destruction or vascular tumors like renal cell carcinoma. Enophthalmos (a rare sign) can be the first sign of orbital scirrhous breast carcinoma. Lateral orbit is the most common location of orbital metastasis (39%), followed by the superior (32%), medial (20%), and inferior (12%) quadrants [106].

On CT scan, metastatic lesion is a solitary mass with irregular borders which may be present with bone destruction, intracranial, or sinus extension [118]. The bony lesions present in prostate carcinoma may be either hyperostotic (osteoblastic) or hypostotic (osteolytic).

Specific tests include affinity for uptake of the radiopharmaceutical metaiodobenzylguanidine (MIBG) in carcinoid tumors [119]. A FNAB or FACT can be used with preparation and evaluation of the specimens by an experienced cytopathologist. An incisional biopsy is indicated where FNAB results are inconclusive or when it cannot be performed because of location. Currently, immunohistochemical staining techniques are the gold standard to resolve most differential diagnosis using tissue-specific antibodies.

Management of orbital metastatic disease includes surgical excision, palliative irradiation, hormone therapy, chemotherapy, and observation depending on clinical circumstances. Surgical excision is rarely done and only for a solitary, well-circumscribed orbital mass.

External beam radiotherapy doses are divided over periods of 2 to 4 weeks for a total dose of 30 to 50 Gy (also recommended in daily 180–200 cGy fractions) [120]. In advanced cases of orbital metastases, which may also involve adjacent structures, high-dose radiation (HDR) brachytherapy may be used for palliative purposes [120].

Hormonal therapy (tamoxifen) is an alternative to more aggressive methods of treatment in tumors that show estrogen and progesterone surface receptors like breast and prostate carcinoma [121].

Chemotherapy is indicated in tumors that are unresponsive to hormonal therapy and in premenopausal women. Treatment of metastatic breast cancer will usually involve hormone therapy and/or chemotherapy with or without trastuzumab (Herceptin) in patients whose tumors overexpress HER2-neu receptors [122].

In patients with prostate carcinoma, the diagnosis of metastatic orbital disease is facilitated by the demonstration of osteoblastic and/or osteolytic bone lesions associated with pain. Confirmation is provided by the measurement of prostate-specific antigen (PSA) in serum or the expression of the same antigen in tissue sections from the metastatic tumor [123, 124].

Observation without specific treatment of the orbital tumor is a choice for patients with asymptomatic orbital metastasis with pre-morbid, advanced systemic disease. The prognosis of patients with orbital metastases is poor. Authors have reported that 95% patients have died of metastasis, with overall mean survival of 15 months, ranging from 3 to 96 months after diagnosis [102, 109].

The information provided in this review reflects the recent advances about the most common ocular malignant lesions. We hope that this review will benefit any clinician dealing with ocular malignancies and any researcher interested in the challenge of curing what are believed to be difficult tumors with high mortality and significant morbidity.

Funding This study was supported by The Eye Cancer Foundation, Inc. Dr. Maheshwari received a fellowship grant from The Eye Cancer Foundation (<http://eyecancercure.com>).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Margo, C. E., & Waltz, K. (1993). Basal cell carcinoma of the eyelid and periocular skin. *Survey of Ophthalmology*, 38(2), 169–192.
2. Esmaeli, B., Dutton, J. J., Graue, G. F., et al. (2016). Eyelid carcinoma. In M. B. Amin, S. B. Edge, F. L. Greene, et al. (Eds.), *AJCC cancer staging manual* (pp. 779–785). New York: Springer International Publishing.
3. Bakis, S., Irwig, L., Wood, G., & Wong, D. (2004). Exfoliative cytology as a diagnostic test for basal cell carcinoma: a meta-analysis. *The British Journal of Dermatology*, 150(5), 829–836.
4. Cook, B. E., & Bartley, G. B. (2001). Treatment options and future prospects for the management of eyelid malignancies: an evidence-based update. *Ophthalmology*, 108(11), 2088–2098.

5. Kokoszka, A., & Scheinfeld, N. (2003). Evidence-based review of the use of cryosurgery in treatment of basal cell carcinoma. *Dermatologic Surgery: Official Publication for American Society for Dermatologic Surgery [et al.]*, 29(6), 566–571.
6. Lindgren, G., & Larkö, O. (1997). Long-term follow-up of cryosurgery of basal cell carcinoma of the eyelid. *Journal of the American Academy of Dermatology*, 36(5 Pt 1), 742–746.
7. Tuppurainen, K. (1995). Cryotherapy for eyelid and periocular basal cell carcinomas: outcome in 166 cases over an 8-year period. *Graefes Archive for Clinical and Experimental Ophthalmology = Albrecht von Graefes Archiv für Klinische und Experimentelle Ophthalmologie*, 233(4), 205–208.
8. Hughes, W. L. (1986). Reconstructive surgery of the eyelids. *Advances in Ophthalmic Plastic and Reconstructive Surgery*, 5, 25–87.
9. Kopecky, A., Koch, K. R., Bucher, F., Cursiefen, C., & Heindl, L. M. (2016). Results of Cutler-Beard procedure for reconstruction of extensive full thickness upper eyelid defects following tumor resection. *Der Ophthalmologe: Zeitschrift der Deutschen Ophthalmologischen Gesellschaft*, 113(4), 309–313.
10. Belmajdoub, M., Jacomet, P.-V., Benillouche, P., & Galatoire, O. (2015). Upper eyelid reconstruction with the Cutler-Beard flap technique: retrospective study of 16 cases. *Journal Français d'Ophthalmologie*, 38(7), 607–614.
11. Leibovitch, I., McNab, A., Sullivan, T., Davis, G., & Selva, D. (2005). Orbital invasion by periocular basal cell carcinoma. *Ophthalmology*, 112(4), 717–723.
12. Karabulut, G. O., Kaynak, P., Ozturker, C., Fazil, K., Ocak, O. B., & Taskapılı, M. (2017). Imiquimod 5% cream for the treatment of large nodular basal cell carcinoma at the medial canthal area. *Indian Journal of Ophthalmology*, 65(1), 48–51.
13. de Macedo, E. M. S., Carneiro, R. C., de Lima, P. P., Silva, B. G., & Matayoshi, S. (2015). Imiquimod cream efficacy in the treatment of periocular nodular basal cell carcinoma: a non-randomized trial. *BMC Ophthalmology*, 15, 35.
14. Macha, M. A., Batra, S. K., & Ganti, A. K. (2013). Profile of vismodegib and its potential in the treatment of advanced basal cell carcinoma. *Cancer Management and Research*, 5, 197–203.
15. Ozgur, O. K., Yin, V., Chou, E., Ball, S., Kies, M., William, W. N., Migden, M., Thuro, B. A., & Esmali, B. (2015). Hedgehog pathway inhibition for locally advanced periocular basal cell carcinoma and basal cell nevus syndrome. *American Journal of Ophthalmology*, 160(2), 220–227.
16. Jefford, M., Kiffer, J. D., Somers, G., Daniel, F. J., & Davis, I. D. (2004). Metastatic basal cell carcinoma: rapid symptomatic response to cisplatin and paclitaxel. *ANZ Journal of Surgery*, 74(8), 704–705.
17. Lee, G. A., & Hirst, L. W. (1995). Ocular surface squamous neoplasia. *Survey of Ophthalmology*, 39(6), 429–450.
18. Sun, E. C., Fears, T. R., & Goedert, J. J. (1997). Epidemiology of squamous cell conjunctival cancer. *Cancer Epidemiology, Biomarkers & Prevention: A Publication of the American Association for Cancer Research, cosponsored by the American Society of Preventive Oncology*, 6(2), 73–77.
19. Lee, G. A., & Hirst, L. W. (1992). Incidence of ocular surface epithelial dysplasia in metropolitan Brisbane. A 10-year survey. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 110(4), 525–527.
20. Char, D. H., Crawford, J. B., Howes, E. L., & Weinstein, A. J. (1992). Resection of intraocular squamous cell carcinoma. *The British Journal of Ophthalmology*, 76(2), 123–125.
21. Semenova, E. A., Milman, T., Finger, P. T., Natesh, S., Kurli, M., Schneider, S., Jacob, C. E., & McCormick, S. A. (2009). The diagnostic value of exfoliative cytology vs histopathology for ocular surface squamous neoplasia. *American Journal of Ophthalmology*, 148(5), 772–778.
22. Finger, P. T., Tran, H. V., Turbin, R. E., et al. (2003). High-frequency ultrasonographic evaluation of conjunctival intraepithelial neoplasia and squamous cell carcinoma. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 121(2), 168–172.
23. Conway, R. M., Graue, G. F., Pelayes, D., et al. (2016). Conjunctival carcinoma. In M. B. Amin, S. B. Edge, F. L. Greene, et al. (Eds.), *AJCC cancer staging manual* (pp. 787–793). New York: Springer International Publishing.
24. Finger, P. T. (2005). “Finger-tip” cryotherapy probes: treatment of squamous and melanocytic conjunctival neoplasia. *The British Journal of Ophthalmology*, 89(8), 942–945.
25. Yousef, Y. A., & Finger, P. T. (2012). Squamous carcinoma and dysplasia of the conjunctiva and cornea: an analysis of 101 cases. *Ophthalmology*, 119(2), 233–240.
26. Chaugule, S. S., Park, J., & Finger, P. T. (2018). Topical chemotherapy for giant ocular surface squamous neoplasia of the conjunctiva and cornea: is surgery necessary? *Indian Journal of Ophthalmology*, 66(1), 55.
27. Karp, C. L., Galor, A., Chhabra, S., Barnes, S. D., & Alfonso, E. C. (2010). Subconjunctival/perilesional recombinant interferon $\alpha 2b$ for ocular surface squamous neoplasia: a 10-year review. *Ophthalmology*, 117(12), 2241–2246.
28. Graue, G. F., Tena, L. B., & Finger, P. T. (2011). Electron beam radiation for conjunctival squamous carcinoma. *Ophthalmic Plastic and Reconstructive Surgery*, 27(4), 277–281.
29. Teng, C. C., Chin, K. J., & Finger, P. T. (2009). Subconjunctival ranibizumab for squamous cell carcinoma of the conjunctiva with corneal extension. *The British Journal of Ophthalmology*, 93(6), 837–838.
30. Finger, P. T. (2009). Radiation therapy for orbital tumors: concepts, current use, and ophthalmic radiation side effects. *Survey of Ophthalmology*, 54(5), 545–568.
31. Fraunfelder, F. T., Wallace, T. R., Farris, H. E., et al. (1977). The role of cryosurgery in external ocular and periocular disease. *Transactions. Section of Ophthalmology. American Academy of Ophthalmology and Otolaryngology*, 83(4 Pt 1), 713–724.
32. Erie, J. C., Campbell, R. J., & Liesegang, T. J. (1986). Conjunctival and corneal intraepithelial and invasive neoplasia. *Ophthalmology*, 93(2), 176–183.
33. Iliff, W. J., Marback, R., & Green, W. R. (1975). Invasive squamous cell carcinoma of the conjunctiva. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 93(2), 119–122.
34. Dimaris, H., Corson, T. W. (2018) Retinoblastoma, the visible CNS tumor: A review. *Journal of Neuroscience Research*. <https://doi.org/10.1002/jnr.24213>.
35. Bishop, J. O., & Madson, E. C. (1975). Retinoblastoma. Review of the current status. *Survey of Ophthalmology*, 19(6), 342–366.
36. Abramson, D. H., Frank, C. M., Susman, M., Whalen, M. P., Dunkel, I. J., & Boyd, N. W. (1998). Presenting signs of retinoblastoma. *The Journal of Pediatrics*, 132(3 Pt 1), 505–508.
37. Ainbinder, D. J., Haik, B. G., Frei, D. F., Gupta, K. L., & Mafee, M. F. (1996). Gadolinium enhancement: improved MRI detection of retinoblastoma extension into the optic nerve. *Neuroradiology*, 38(8), 778–781.
38. Shields, C. L., Fulco, E. M., Arias, J. D., et al. (2013). Retinoblastoma frontiers with intravenous, intra-arterial, periocular, and intravitreal chemotherapy. *Eye (London, England)*, 27(2), 253–264.
39. Mallipatna, A. C., Gallie, B. L., Chevez-Barrios, P., et al. (2016). Retinoblastoma. In M. B. Amin, S. B. Edge, F. L. Greene, et al. (Eds.), *AJCC cancer staging manual* (pp. 819–831). New York: Springer International Publishing.
40. Shields, C. L., Schoenberg, E., Kocher, K., Shukla, S. Y., Kaliki, S., & Shields, J. A. (2013). Lesions simulating retinoblastoma (pseudoretinoblastoma) in 604 cases: results based on age at presentation. *Ophthalmology*, 120(2), 311–316.

41. Karcioğlu, Z. A., Gordon, R. A., & Karcioğlu, G. L. (1985). Tumor seeding in ocular fine needle aspiration biopsy. *Ophthalmology*, 92(12), 1763–1767.
42. Shields, C. L., Shields, J. A., Cater, J., Othmane, I., Singh, A. D., & Micaily, B. (2001). Plaque radiotherapy for retinoblastoma: long-term tumor control and treatment complications in 208 tumors. *Ophthalmology*, 108(11), 2116–2121.
43. Hernandez, J. C., Brady, L. W., Shields, J. A., Shields, C. L., DePotter, P., Karlsson, U. L., Markoe, A. M., Amendola, B. E., & Singh, A. (1996). External beam radiation for retinoblastoma: results, patterns of failure, and a proposal for treatment guidelines. *International Journal of Radiation Oncology, Biology, Physics*, 35(1), 125–132.
44. Honavar, S. G., & Singh, A. D. (2005). Management of advanced retinoblastoma. *Ophthalmology Clinics of North America*, 18(1), 65–73.
45. Munier, F. L., Gaillard, M.-C., Balmer, A., Soliman, S., Podilsky, G., Moulin, A. P., & Beck-Popovic, M. (2012). Intravitreal chemotherapy for vitreous disease in retinoblastoma revisited: from prohibition to conditional indications. *The British Journal of Ophthalmology*, 96(8), 1078–1083.
46. Shields, C. L., Santos, M. C., Diniz, W., et al. (1999). Thermotherapy for retinoblastoma. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 117(7), 885–893.
47. American Brachytherapy Society - Ophthalmic Oncology Task Force., ABS – OOTF Committee. (2014). The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma and retinoblastoma. *Brachytherapy*, 13(1), 1–14.
48. Abramson, D. H., & Frank, C. M. (1998). Second nonocular tumors in survivors of bilateral retinoblastoma: a possible age effect on radiation-related risk. *Ophthalmology*, 105(4), 573–579 discussion 579–580.
49. Murthy, R., Honavar, S. G., Naik, M. N., & Reddy, V. A. P. (2005). Retinoblastoma. In L. Dutta (Ed.), *Modern ophthalmology* (Vol. 2, pp. 849–859). New Delhi: Jaypee Brothers Medical Publishers (P) Ltd.
50. Gupta, R., Vemuganti, G. K., Reddy, V. A. P., & Honavar, S. G. (2009). Histopathologic risk factors in retinoblastoma in India. *Archives of Pathology & Laboratory Medicine*, 133(8), 1210–1214.
51. Finger, P. T., Harbour, J. W., & Karcioğlu, Z. A. (2002). Risk factors for metastasis in retinoblastoma. *Survey of Ophthalmology*, 47(1), 1–16.
52. Gichigo, E. N., Kariuki-Wanyoike, M. M., Kimani, K., & Nentwich, M. M. (2015). Retinoblastoma in Kenya: survival and prognostic factors. *Der Ophthalmologe: Zeitschrift der Deutschen Ophthalmologischen Gesellschaft*, 112(3), 255–260.
53. Singh, A. D., Bergman, L., & Seregard, S. (2005). Uveal melanoma: epidemiologic aspects. *Ophthalmology Clinics of North America*, 18(1), 75–84 viii.
54. Hu, D.-N., Yu, G.-P., McCormick, S. A., Schneider, S., & Finger, P. T. (2005). Population-based incidence of uveal melanoma in various races and ethnic groups. *American Journal of Ophthalmology*, 140(4), 612–617.
55. Hawkins, B. S., & Collaborative Ocular Melanoma Study Group. (2004). The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma: IV. Ten-year mortality findings and prognostic factors. COMS report number 24. *American Journal of Ophthalmology*, 138(6), 936–951.
56. Inskip, P. D., Devesa, S. S., & Fraumeni, J. F. (2003). Trends in the incidence of ocular melanoma in the United States, 1974–1998. *Cancer Causes Control CCC*, 14(3), 251–257.
57. Singh, A. D., & Topham, A. (2003). Incidence of uveal melanoma in the United States: 1973–1997. *Ophthalmology*, 110(5), 956–961.
58. Guénel, P., Laforest, L., Cyr, D., Fèveotte, J., Sabroe, S., Dufour, C., Lutz, J. M., & Lynge, E. (2001). Occupational risk factors, ultraviolet radiation, and ocular melanoma: a case-control study in France. *Cancer Causes Control CCC*, 12(5), 451–459.
59. Lutz, J.-M., Cree, I., Sabroe, S., Kvist, T. K., Clausen, L. B., Afonso, N., Ahrens, W., Ballard, T. J., Bell, J., Cyr, D., Eriksson, M., Fèveotte, J., Guénel, P., Hardell, L., Jöckel, K. H., Miranda, A., Merletti, F., Morales-Suarez-Varela, M. M., Stengrevics, A., & Lynge, E. (2005). Occupational risks for uveal melanoma results from a case-control study in nine European countries. *Cancer Causes Control CC*, 16(4), 437–447.
60. Vajdic, C. M., Kricker, A., Giblin, M., McKenzie, J., Aitken, J., Giles, G. G., & Armstrong, B. K. (2002). Sun exposure predicts risk of ocular melanoma in Australia. *International Journal of Cancer*, 101(2), 175–182.
61. Finger, P. T. (2015). Ocular malignancies: choroidal melanoma, retinoblastoma, ocular adnexal lymphoma and eyelid cancer. In B. O’Sullivan & S. H. Huang (Eds.), *UICC manual of clinical oncology* (pp. 726–745). West Sussex: Wiley.
62. Lemaître, S., Zmuda, M., Jacomet, P. V., Lévy-Gabriel, C., Dendale, R., Berges, O., Fréneaux, P., Mariani, P., Desjardins, L., Galatoire, O., & Cassoux, N. (2017). Small choroidal melanoma revealed by a large extrascleral extension. *Ocular Oncology and Pathology*, 3(3), 240–246.
63. Lorenzano, D., Miszkiel, K., & Rose, G. E. (2017). Orbital melanoma masquerading as a “galloping hamangioma.”. *Orbit (Amsterdam, Netherlands)*, 36(2), 81–83.
64. Freton, A., Chin, K. J., Raut, R., Tena, L. B., Kivelä, T., Finger, P. T. (2002). Initial PET/CT staging for choroidal melanoma: AJCC correlation and second nonocular primaries in 333 patients. *European Journal of Ophthalmology*, 22(2), 236–243.
65. Augsburger, J. J., & Shields, J. A. (1984). Fine needle aspiration biopsy of solid intraocular tumors: indications, instrumentation and techniques. *Ophthalmic Surgery*, 15(1), 34–40.
66. Kivela, T., Simpson, E. R., Grossniklaus, H. E., et al. (2016). Uveal melanoma. In M. B. Amin, S. B. Edge, F. L. Greene, et al. (Eds.), *AJCC cancer staging manual* (pp. 805–813). New York: Springer International Publishing.
67. Earle, J., Kline, R. W., & Robertson, D. M. (1987). Selection of iodine 125 for the Collaborative Ocular Melanoma Study. *Archives of Ophthalmology*, 105(6), 763–764.
68. De Potter, P., Shields, C. L., Shields, J. A., Cater, J. R., & Brady, L. W. (1996). Plaque radiotherapy for juxtapapillary choroidal melanoma. Visual acuity and survival outcome. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 114(11), 1357–1365.
69. Finger, P. T., Chin, K. J., & Duvall, G. (2009). Palladium-103 for Choroidal Melanoma Study Group. Palladium-103 ophthalmic plaque radiation therapy for choroidal melanoma: 400 treated patients. *Ophthalmology*, 116(4), 790–796.
70. Maheshwari, A., Finger, P. T. (2018) A 12-year study of slotted palladium-103 plaque radiation therapy for choroidal melanoma: near, touching, or surrounding the optic nerve. *American Journal of Ophthalmology*, 188, 60–69.
71. Chiu-Tsao, S.-T., Astrahan, M. A., Finger, P. T., et al. (2012). Dosimetry of (125)I and (103)Pd COMS eye plaques for intraocular tumors: report of Task Group 129 by the AAPM and ABS. *Medical Physics*, 39(10), 6161–6184.
72. Shields, C. L., Shields, J. A., Cater, J., et al. (2000). Plaque radiotherapy for uveal melanoma: long-term visual outcome in 1106 consecutive patients. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 118(9), 1219–1228.

73. Finger, P. T., Lu, D., Buffa, A., DeBlasio, D. S., & Bosworth, J. L. (1993). Palladium-103 versus iodine-125 for ophthalmic plaque radiotherapy. *International Journal of Radiation Oncology, Biology, Physics*, 27(4), 849–854.
74. Gragoudas, E. S., Seddon, J. M., Egan, K., Glynn, R., Munzenrider, J., Austin-Seymour, M., Goitein, M., Verhey, L., Urie, M., & Koehler, A. (1987). Long-term results of proton beam irradiated uveal melanomas. *Ophthalmology*, 94(4), 349–353.
75. Krema, H., Somani, S., Sahgal, A., Xu, W., Heydarian, M., Payne, D., McGowan, H., Michaels, H., Simpson, E. R., & Laperriere, N. (2009). Stereotactic radiotherapy for treatment of juxtapapillary choroidal melanoma: 3-year follow-up. *The British Journal of Ophthalmology*, 93(9), 1172–1176.
76. Char, D. H., Bove, R., & Phillips, T. L. (2003). Laser and proton radiation to reduce uveal melanoma-associated exudative retinal detachments. *American Journal of Ophthalmology*, 136(1), 180–182.
77. Wachtlin, J., Bechrakis, N. E., & Foerster, M. H. (2005). Photodynamic therapy with verteporfin for uveal melanoma. *Der Ophthalmologe: Zeitschrift der Deutschen Ophthalmologischen Gesellschaft*, 102(3), 241–246.
78. Wilson, D. J., & Klein, M. L. (2002). Choroidal melanoma treated with cryotherapy. *Archives of Ophthalmology*, 120(3), 393–395.
79. Augsburger, J. J., & Gamel, J. W. (1990). Clinical prognostic factors in patients with posterior uveal malignant melanoma. *Cancer*, 66(7), 1596–1600.
80. Collaborative Ocular Melanoma Study Group. (2001). Assessment of metastatic disease status at death in 435 patients with large choroidal melanoma in the Collaborative Ocular Melanoma Study (COMS): COMS report no. 15. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 119(5), 670–676.
81. Sjö, L. D., Ralfkiaer, E., Prause, J. U., et al. (2008). Increasing incidence of ophthalmic lymphoma in Denmark from 1980 to 2005. *Investigative Ophthalmology & Visual Science*, 49(8), 3283–3288.
82. Coupland, S. E., Hellmich, M., Auw-Haendrich, C., Lee, W. R., & Stein, H. (2004). Prognostic value of cell-cycle markers in ocular adnexal lymphoma: an assessment of 230 cases. *Graefes Archive for Clinical and Experimental Ophthalmology = Albrecht von Graefes Archiv für Klinische und Experimentelle Ophthalmologie*, 242(2), 130–145.
83. Knowles, D. M., Jakobiec, F. A., McNally, L., & Burke, J. S. (1990). Lymphoid hyperplasia and malignant lymphoma occurring in the ocular adnexa (orbit, conjunctiva, and eyelids): a prospective multiparametric analysis of 108 cases during 1977 to 1987. *Human Pathology*, 21(9), 959–973.
84. Shields, C. L., Shields, J. A., Carvalho, C., Rundle, P., & Smith, A. F. (2001). Conjunctival lymphoid tumors: clinical analysis of 117 cases and relationship to systemic lymphoma. *Ophthalmology*, 108(5), 979–984.
85. Heegaard, S., Chevez-Barrios, P., White, V. A., Coupland, S. E., & Finger, P. T. (2016). Ocular adnexal lymphoma. In M. B. Amin, S. B. Edge, F. L. Greene, et al. (Eds.), *AJCC cancer staging manual* (pp. 849–854). New York: Springer International Publishing.
86. Hepler, R. S. (1972). Clinical diagnosis of orbital tumors. *Radiologic Clinics of North America*, 10(1), 3–10.
87. Gerbino, G., Boffano, P., Benech, R., et al. (2014). Orbital lymphomas: clinical and radiological features. *Journal of Cranio-Maxillo-Facial Surgery: Official Publication of the European Association for Cranio-Maxillo-Facial Surgery*, 42(5), 508–512.
88. Coupland, S. E., Krause, L., Delecluse, H. J., et al. (1998). Lymphoproliferative lesions of the ocular adnexa. Analysis of 112 cases. *Ophthalmology*, 105(8), 1430–1441.
89. English, J. F., & Sullivan, T. J. (2015). The role of FDG-PET in the diagnosis and staging of ocular adnexal lymphoproliferative disease. *Orbit (Amsterdam, Netherlands)*, 34(5), 284–291.
90. Hui, K.-H., Pfeiffer, M. L., & Esmali, B. (2012). Value of positron emission tomography/computed tomography in diagnosis and staging of primary ocular and orbital tumors. *Saudi Journal of Ophthalmology: Official Journal of the Saudi Ophthalmological Society*, 26(4), 365–371.
91. Fung, C. Y., Tarbell, N. J., Lucarelli, M. J., Goldberg, S. I., Linggood, R. M., Harris, N. L., & Ferry, J. A. (2003). Ocular adnexal lymphoma: clinical behavior of distinct World Health Organization classification subtypes. *International Journal of Radiation Oncology, Biology, Physics*, 57(5), 1382–1391.
92. Leget, G. A., & Czuczman, M. S. (1998). Use of rituximab, the new FDA-approved antibody. *Current Opinion in Oncology*, 10(6), 548–551.
93. Maloney, D. G., Grillo-López, A. J., White, C. A., et al. (1997). IDEC-C2B8 (rituximab) anti-CD20 monoclonal antibody therapy in patients with relapsed low-grade non-Hodgkin's lymphoma. *Blood*, 90(6), 2188–2195.
94. McLaughlin, P., Grillo-López, A. J., Link, B. K., Levy, R., Czuczman, M. S., Williams, M. E., Heyman, M. R., Bence-Bruckler, I., White, C. A., Cabanillas, F., Jain, V., Ho, A. D., Lister, J., Wey, K., Shen, D., & Dallaire, B. K. (1998). Rituximab chimeric anti-CD20 monoclonal antibody therapy for relapsed indolent lymphoma: half of patients respond to a four-dose treatment program. *Journal of Clinical Oncology: Official Journal of the American Society of Clinical Oncology*, 16(8), 2825–2833.
95. Chan, C.-C., Smith, J. A., Shen, D. F., Ursea, R., LeHoang, P., & Grossniklaus, H. E. (2004). Helicobacter pylori (H. pylori) molecular signature in conjunctival mucosa-associated lymphoid tissue (MALT) lymphoma. *Histology and Histopathology*, 19(4), 1219–1226.
96. Ferreri, A. J. M., Guidoboni, M., Ponzone, M., de Conciliis, C., Dell'Oro, S., Fleischhauer, K., Caggiari, L., Lettini, A. A., Dal Cin, E., Ieri, R., Freschi, M., Villa, E., Boiocchi, M., & Dolcetti, R. (2004). Evidence for an association between Chlamydia psittaci and ocular adnexal lymphomas. *Journal of the National Cancer Institute*, 96(8), 586–594.
97. Kirkegaard, M. M., Rasmussen, P. K., Coupland, S. E., Esmali, B., Finger, P. T., Graue, G. F., Grossniklaus, H. E., Honavar, S. G., Khong, J. J., McKelvie, P. A., Mulay, K., Prause, J. U., Ralfkiaer, E., Sjö, L. D., Toft, P. B., Vemuganti, G. K., Thuro, B. A., Curtin, J., & Heegaard, S. (2016). Conjunctival lymphoma—an international multicenter retrospective study. *JAMA Ophthalmology*, 134(4), 406–414.
98. Rasmussen, P. K., Coupland, S. E., Finger, P. T., Graue, G. F., Grossniklaus, H. E., Honavar, S. G., McKelvie, P., Mulay, K., Prause, J. U., Ralfkiaer, E., Sjö, L. D., & Heegaard, S. (2014). Ocular adnexal follicular lymphoma: a multicenter international study. *JAMA Ophthalmology*, 132(7), 851–858.
99. Svendsen, F. H., Rasmussen, P. K., Coupland, S. E., Esmali, B., Finger, P. T., Graue, G. F., Grossniklaus, H. E., Honavar, S. G., Khong, J. J., McKelvie, P. A., Mulay, K., Ralfkiaer, E., Sjö, L. D., Vemuganti, G. K., Thuro, B. A., Curtin, J., & Heegaard, S. (2017). Lymphoma of the eyelid—an international multicenter retrospective study. *American Journal of Ophthalmology*, 177, 58–68.
100. Jenkins, C., Rose, G. E., Bunce, C., et al. (2000). Histological features of ocular adnexal lymphoma (REAL classification) and their association with patient morbidity and survival. *The British Journal of Ophthalmology*, 84(8), 907–913.
101. McKelvie, P. A., McNab, A., Francis, I. C., Fox, R., & O'Day, J. (2001). Ocular adnexal lymphoproliferative disease: a series of 73 cases. *Clinical & Experimental Ophthalmology*, 29(6), 387–393.
102. Char, D. H., Miller, T., & Kroll, S. (1997). Orbital metastases: diagnosis and course. *The British Journal of Ophthalmology*, 81(5), 386–390.

103. Ferry, A. P., & Font, R. L. (1974). Carcinoma metastatic to the eye and orbit. I. A clinicopathologic study of 227 cases. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 92(4), 276–286.
104. Kennedy, R. E. (1984). An evaluation of 820 orbital cases. *Transactions of the American Ophthalmological Society*, 82, 134–157.
105. Shields, J. A., Bakewell, B., Augsburger, J. J., & Flanagan, J. C. (1984). Classification and incidence of space-occupying lesions of the orbit. A survey of 645 biopsies. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 102(11), 1606–1611.
106. Goldberg, R. A., Rootman, J., & Cline, R. A. (1990). Tumors metastatic to the orbit: a changing picture. *Survey of Ophthalmology*, 35(1), 1–24.
107. Günalp, I., & Gündüz, K. (1995). Metastatic orbital tumors. *Japanese Journal of Ophthalmology*, 39(1), 65–70.
108. Shields, C. L., Shields, J. A., & Peggs, M. (1988). Tumors metastatic to the orbit. *Ophthalmic Plastic and Reconstructive Surgery*, 4(2), 73–80.
109. Shields, J. A., Shields, C. L., Brotman, H. K., Carvalho, C., Perez, N., & Eagle, R. C. (2001). Cancer metastatic to the orbit: the 2000 Robert M. Curtis Lecture. *Ophthalmic Plastic and Reconstructive Surgery*, 17(5), 346–354.
110. Bersani, T. A., Costello, J. J., Mango, C. A., & Streeten, B. W. (1994). Benign approach to a malignant orbital tumor: metastatic renal cell carcinoma. *Ophthalmic Plastic and Reconstructive Surgery*, 10(1), 42–44.
111. Betharia, S. M. (1985). Metastatic orbital carcinoma of thyroid. *Indian Journal of Ophthalmology*, 33(3), 191–193.
112. Drake, E. T., & Dobben, G. D. (1959). Leiomyosarcoma of the uterus with unusual metastases. *Journal of the American Medical Association*, 170(11), 1294–1298.
113. Rush, J. A., Older, J. J., & Richman, A. V. (1981). Testicular seminoma metastatic to the orbit. *American Journal of Ophthalmology*, 91(2), 258–260.
114. Saxena, R. B., Mathur, R. N., & Sonani, S. Z. (1975). Orbital metastasis of mixed parotid tumour. *Indian Journal of Ophthalmology*, 23(2), 23–24.
115. Seretan, E. L. (1981). Metastatic adenocarcinoma from the stomach to the orbit. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 99(8), 1469.
116. Smiley, S. S. (1965). An orbital metastasis from the urinary bladder. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 74(6), 809–810.
117. Stewart, W. B., Newman, N. M., Cavender, J. C., & Spencer, W. H. (1978). Fibrous histiocytoma metastatic to the orbit. *Archives of Ophthalmology (Chicago, Ill.: 1960)*, 96(5), 871–873.
118. Healy, J. F. (1983). Computed tomographic evaluation of metastases to the orbit. *Annals of Ophthalmology*, 15(11), 1026–1029.
119. Hanson, M. W., Schneider, A. M., Enterline, D. S., Feldman, J. M., & Gockerman, J. P. (1998). Iodine-131-metaiodobenzylguanidine uptake in metastatic carcinoid tumor to the orbit. *Journal of Nuclear Medicine: Official Publication, Society of Nuclear Medicine*, 39(4), 647–650.
120. Glassburn, J. R., Klionsky, M., & Brady, L. W. (1984). Radiation therapy for metastatic disease involving the orbit. *American Journal of Clinical Oncology*, 7(2), 145–148.
121. Saitoh, A., Amemiya, T., & Tsuda, N. (1997). Metastasis of breast carcinoma to eyelid and orbit of a postmenopausal woman: good response to tamoxifen therapy. *Ophthalmologica. Journal International D'ophtalmologie. International Journal of Ophthalmology. Zeitschrift Fur Augenheilkunde*, 211(6), 362–366.
122. Kuo, S.-C., Hsiao, S.-C., Chiou, C.-C., Chen, F.-F., & Huang, K.-C. (2008). Metastatic carcinoma of the breast: a case with the unusual presentation of unilateral periorbital edema. *Japanese Journal of Ophthalmology*, 52(4), 305–307.
123. Carriere, V. M., Karciglu, Z. A., Apple, D. J., & Insler, M. S. (1982). A case of prostate carcinoma with bilateral orbital metastases and the review of the literature. *Ophthalmology*, 89(4), 402–406.
124. Kopelman, J. E., & Shorr, N. (1987). A case of prostatic carcinoma metastatic to the orbit diagnosed by fine needle aspiration and immunoperoxidase staining for prostatic specific antigen. *Ophthalmic Surgery*, 18(8), 599–603.